

01

Code

specifications

Cuts applied now:

06/11/2025

mutau				
	ttjets	dy	qcd	data
FASE0	1 tau. 1 muon	1 tau. 1 muon	1 tau. 1 muon	1 tau. 1 muon
Doube_medium				
Iso_Mu24	✓	✓	✓	✓
Ele32_WpTight				
Electron_				
lumi filtering			✓	✓
FASE1				
tau id1	>63	>63	>63	>63
tau id2	>7	>7	>7	>7
tau id3	>1	>1	>1	>1
tau pt	>100	>100	>100	>100
mu id	>=3	>=3	>=3	>=3
mu pt	>35	>35	>35	>35
ele id				
ele pt				
delta R	>0.4	>0.4	>0.4	>0.4
eta	<2.4	<2.4	<2.4	<2.4
charge signs	opposite	opposite	same	opposite
scaling factors	✓	✓		
valid jet and MET info	✓	✓		

Selection cuts

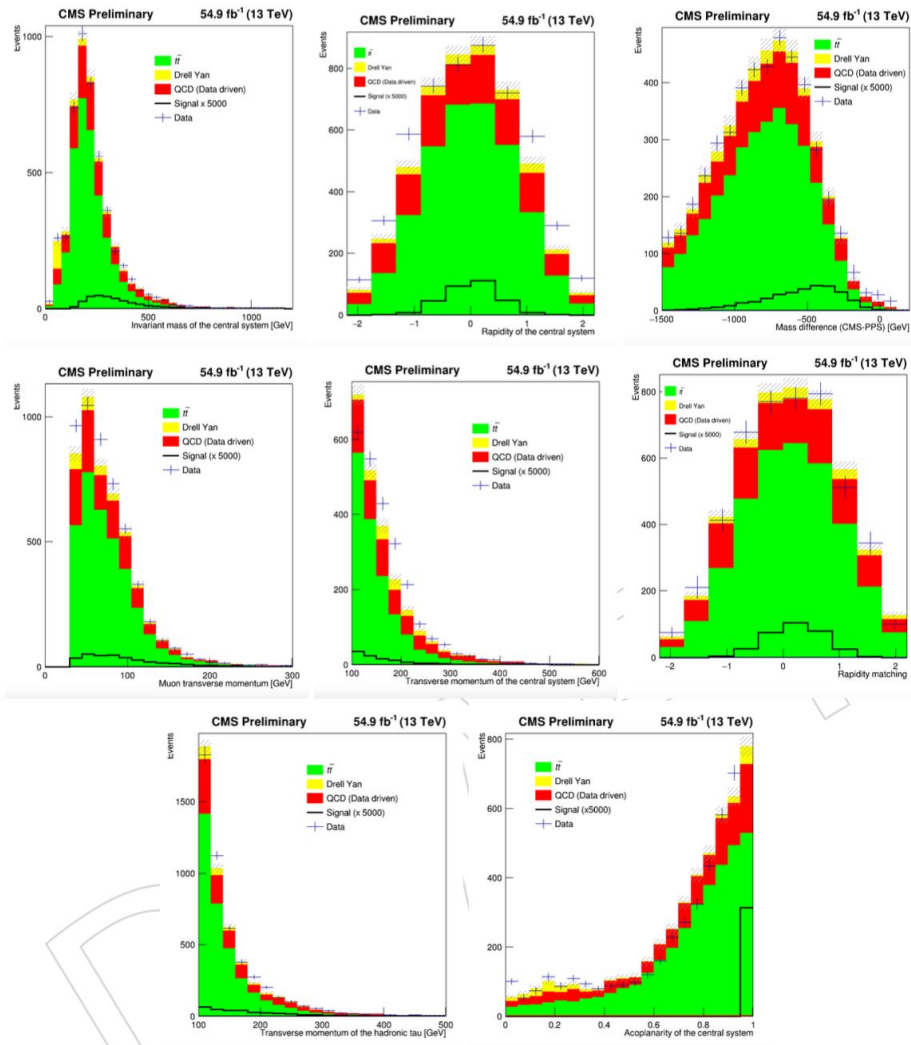
1 muon and 1 τ_h (ID+ISO)
 $p_T(\mu) > 35 \text{ GeV}, p_T(\tau_h) > 100 \text{ GeV}$
 Opposite sign (same-sign for QCD)
 At least two protons in PPS

Cuts explained in the analysis
 note
 proton requirements are applied
 after introducing pileup protons

The issue: Data and QCD samples look smaller than they should

Tried:

- $w_{dy}=1.81$ (fixed), $w_{ttjets}=0..15$ (fixed) and $w_{data}=2$
- $w_{data}=2$
- $w_{data}=2$, $w_{qcd}=2$ -> this gives results similar to matteo's plot



02

Variable plots

Input for the BDT

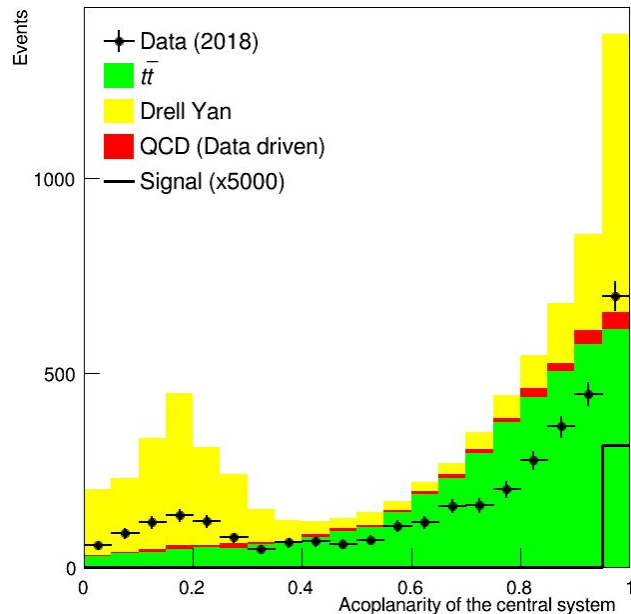
Input for the BDT; using $w_{dy}=1.81$ (fixed),
 $w_{ttjets}=0..15$ (fixed) and $w_{data}=2$

06/11/2025

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

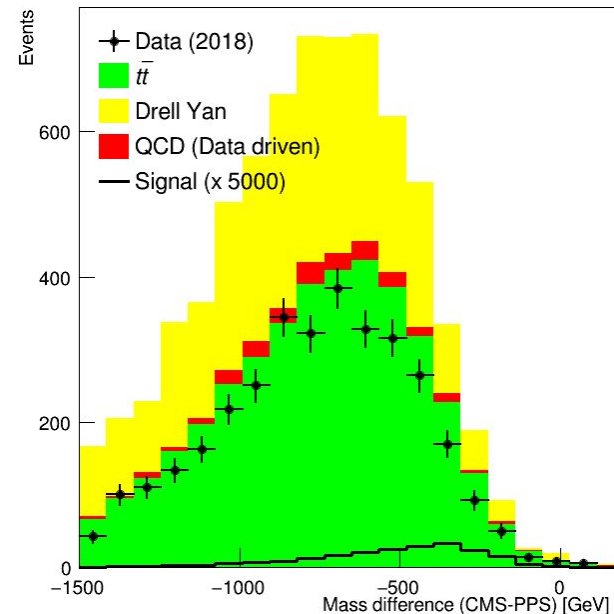
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary



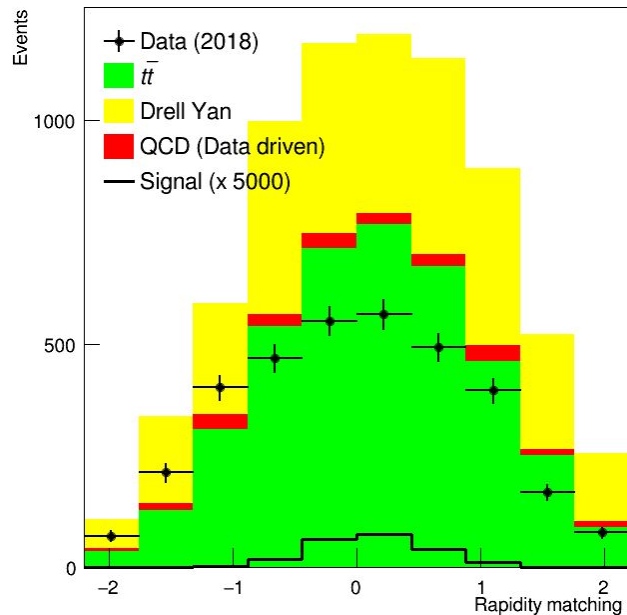
Input for the BDT; using $w_{dy}=1.81$ (fixed),
 $w_{ttjets}=0..15$ (fixed) and $w_{data}=2$

06/11/2025

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

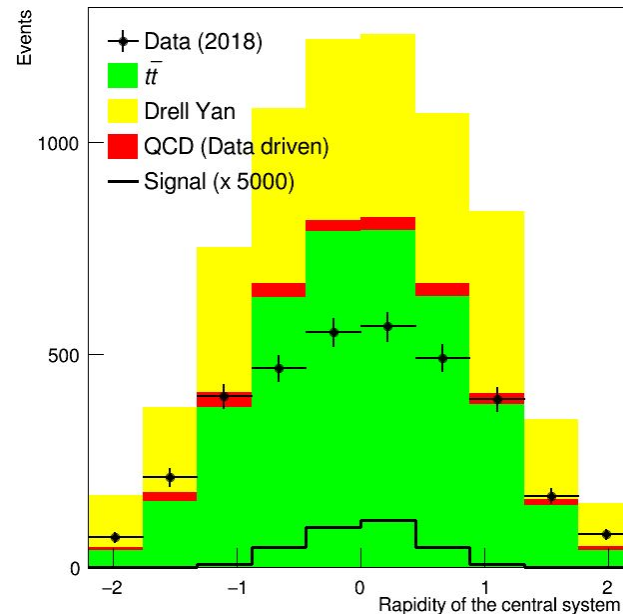
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary



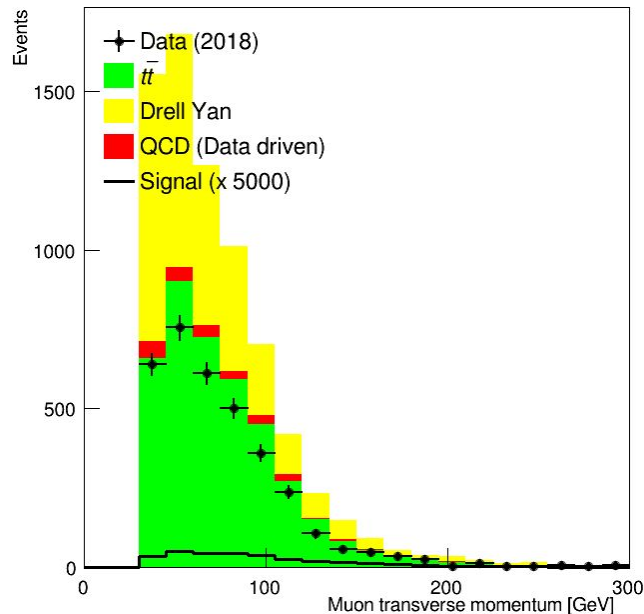
Input for the BDT; using $w_{dy}=1.81$ (fixed),
 $w_{ttjets}=0..15$ (fixed) and $w_{data}=2$

06/11/2025

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

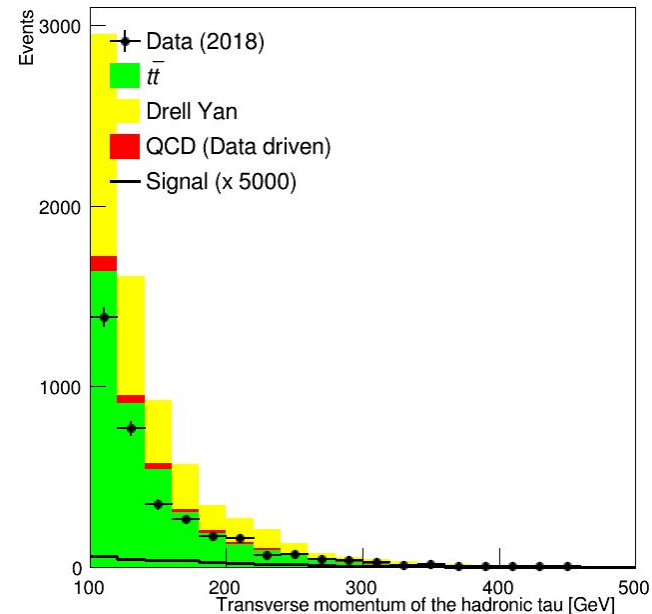
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary



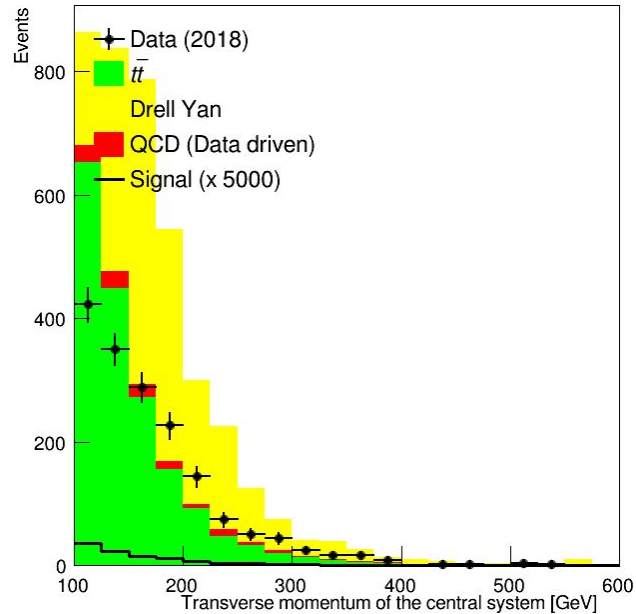
Input for the BDT; using $w_{dy}=1.81$ (fixed),
 $w_{ttjets}=0..15$ (fixed) and $w_{data}=2$

06/11/2025

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

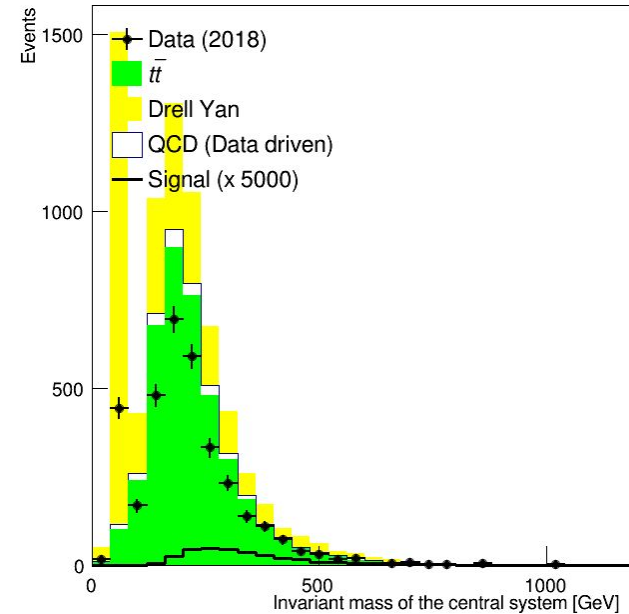
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary

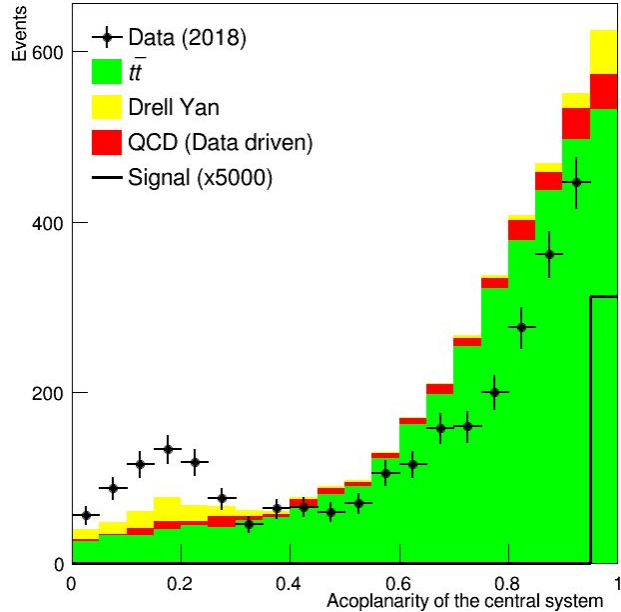


Input for the BDT; using $w_{\text{data}}=2$

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

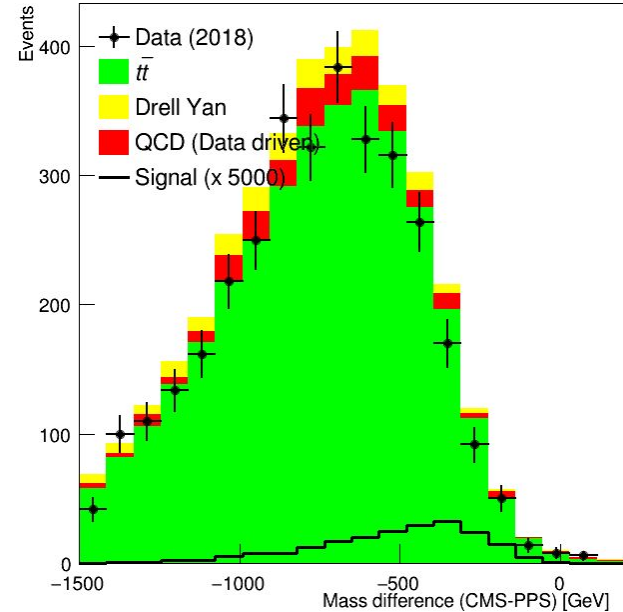
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary

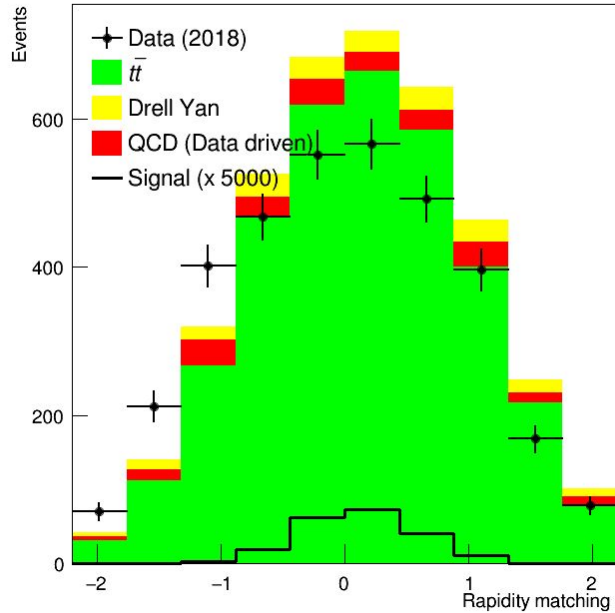


Input for the BDT; using $w_{\text{data}}=2$

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

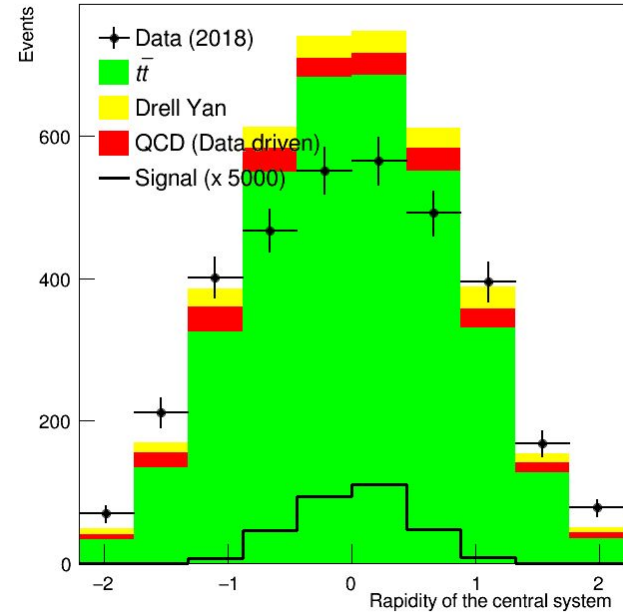
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary

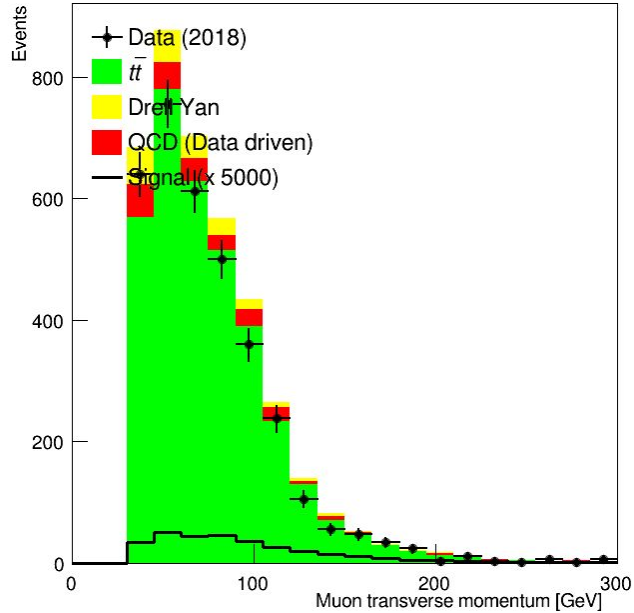


Input for the BDT; using $w_{\text{data}}=2$

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

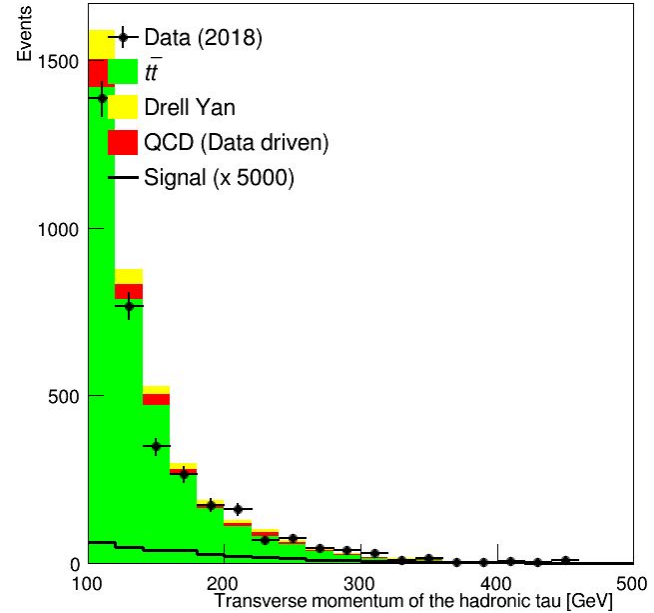
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary

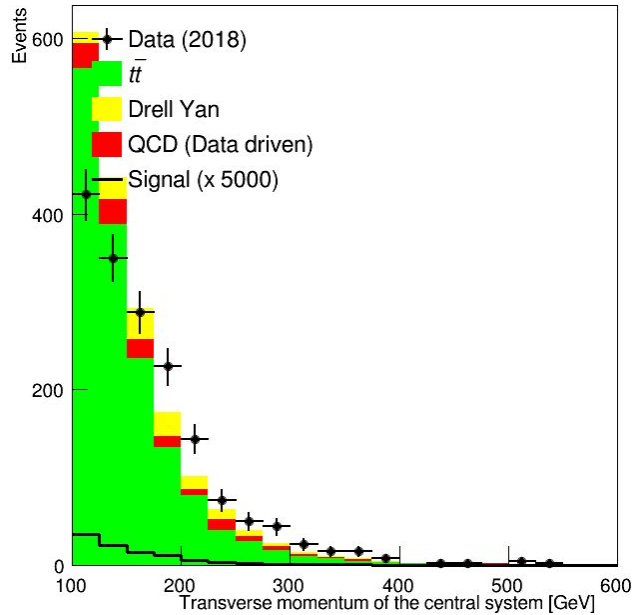


Input for the BDT; using $w_{\text{data}}=2$

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

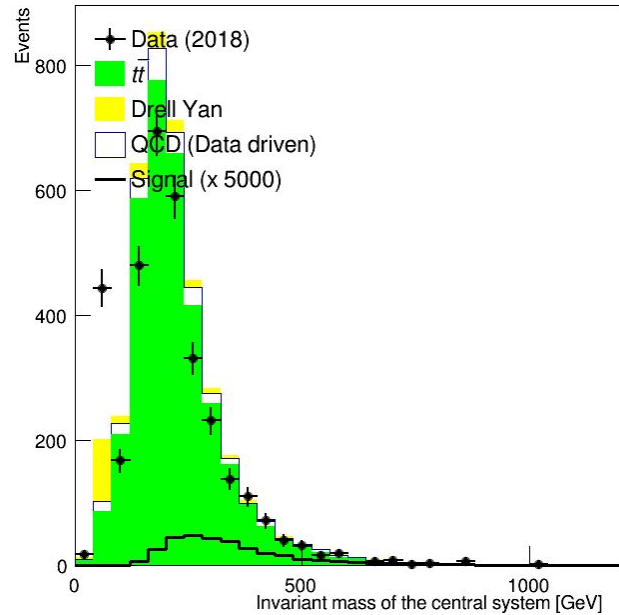
Preliminary



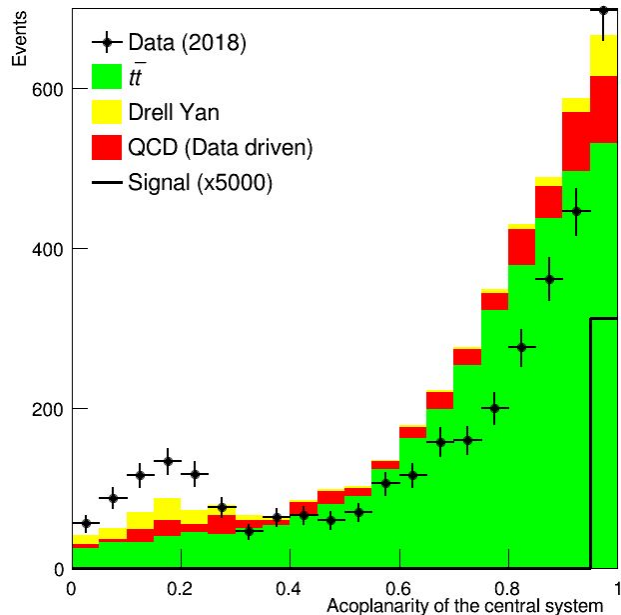
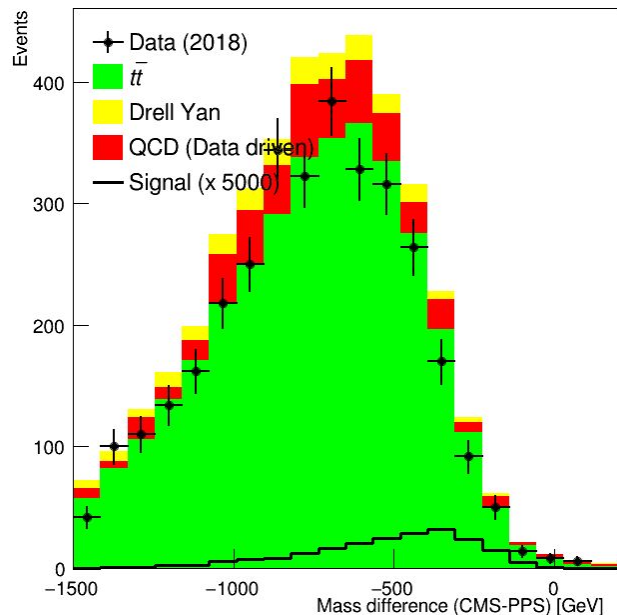
CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary



Input for the BDT; using $w_{\text{data}}=2$, $w_{\text{qcd}}=2$

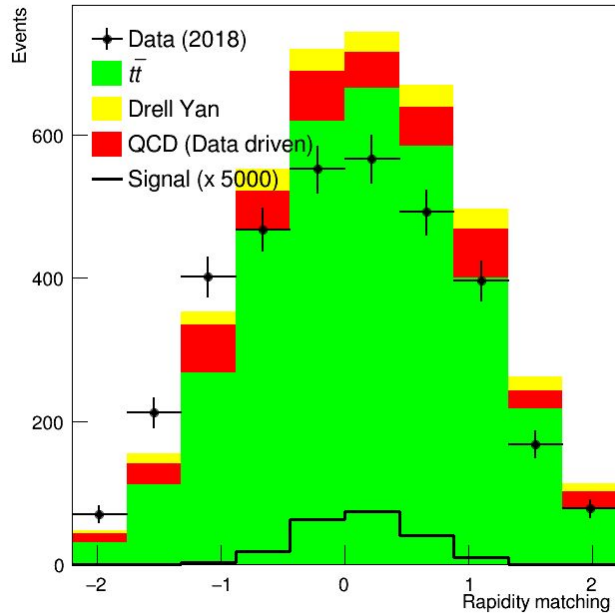
CMS-TOTEM54.9 fb⁻¹ (13 TeV)*Preliminary***CMS-TOTEM**54.9 fb⁻¹ (13 TeV)*Preliminary*

Input for the BDT; using $w_{\text{data}}=2$, $w_{\text{qcd}}=2$

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

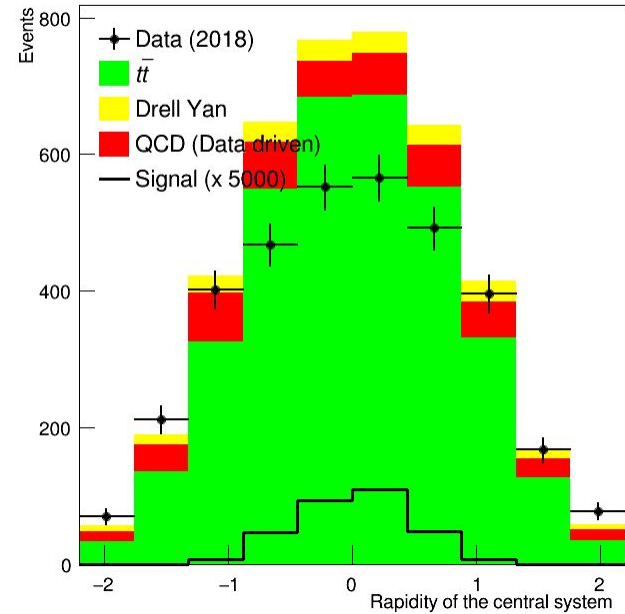
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary

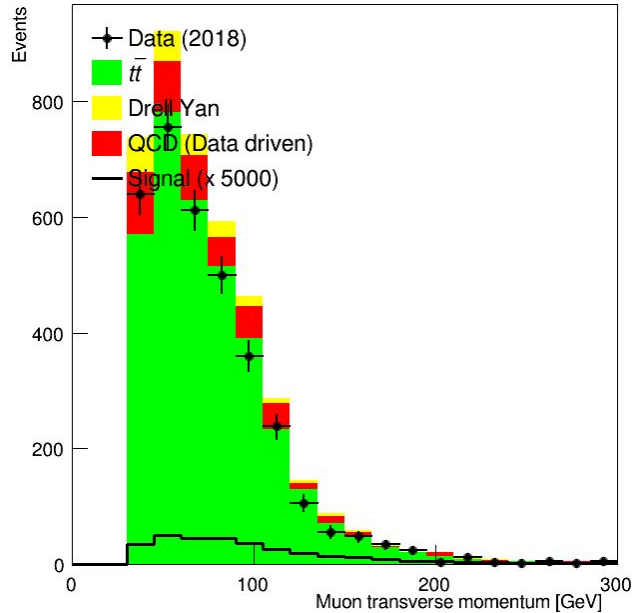


Input for the BDT; using $w_{\text{data}}=2$, $w_{\text{qcd}}=2$

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

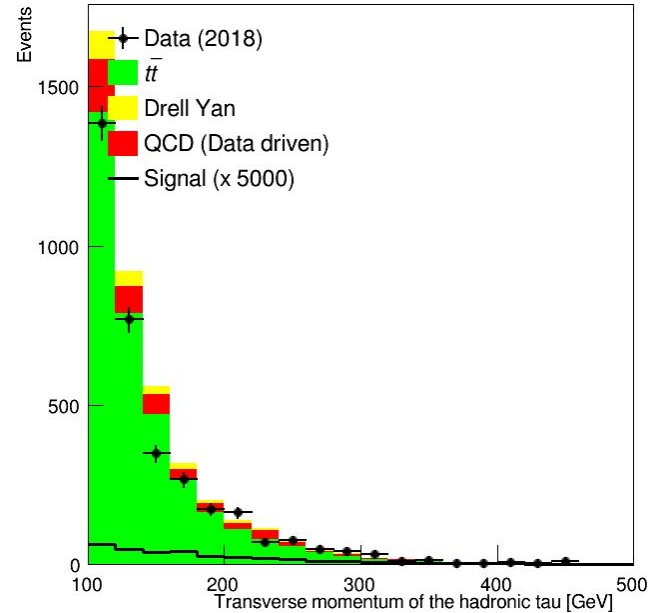
Preliminary



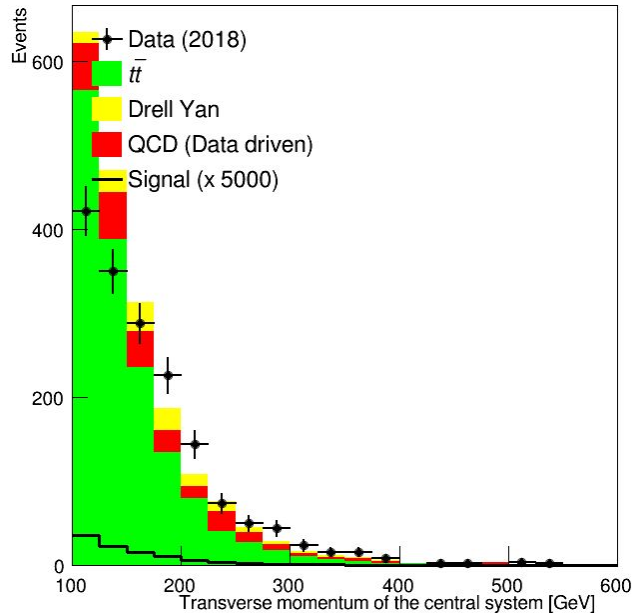
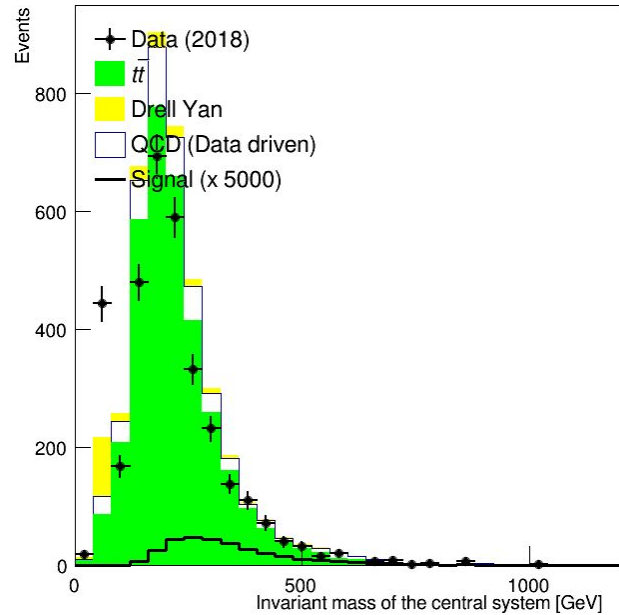
CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary



Input for the BDT; using $w_{\text{data}}=2$, $w_{\text{qcd}}=2$

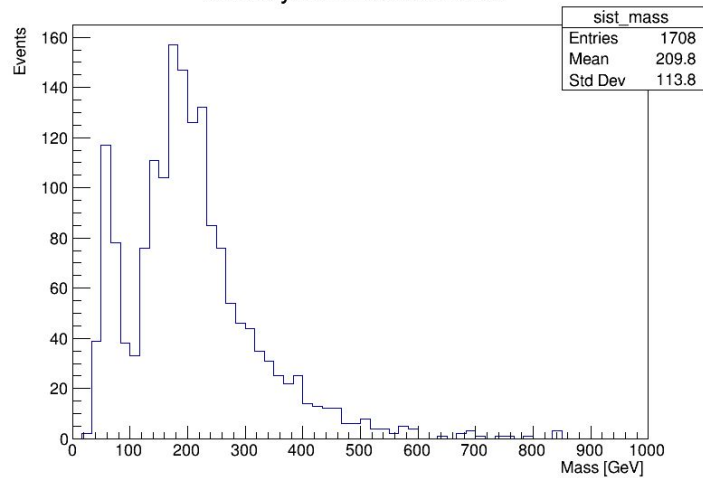
CMS-TOTEM**54.9 fb⁻¹ (13 TeV)***Preliminary***CMS-TOTEM****54.9 fb⁻¹ (13 TeV)***Preliminary*

02

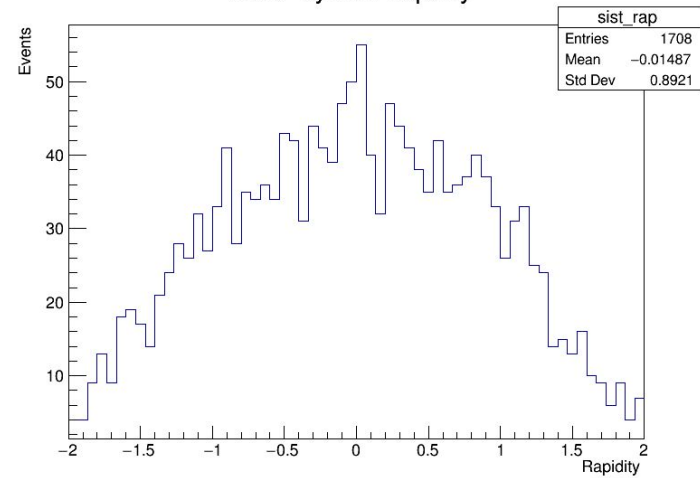
Variable plots

DATA

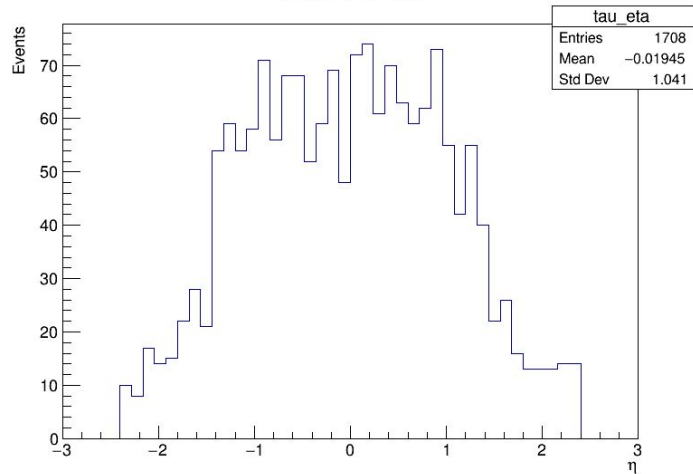
Data: System Invariant Mass



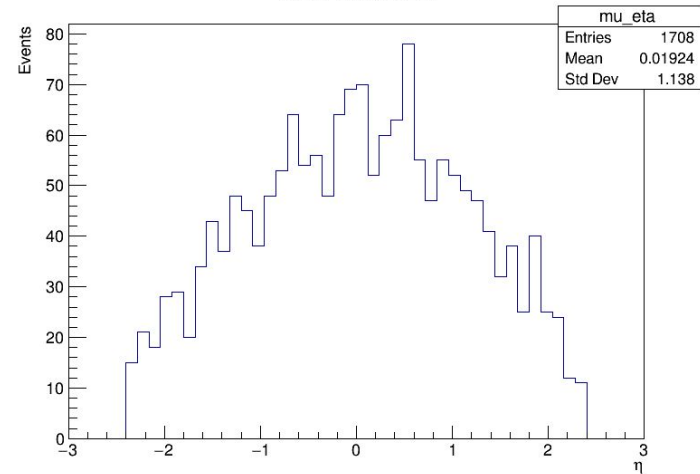
Data: System Rapidity



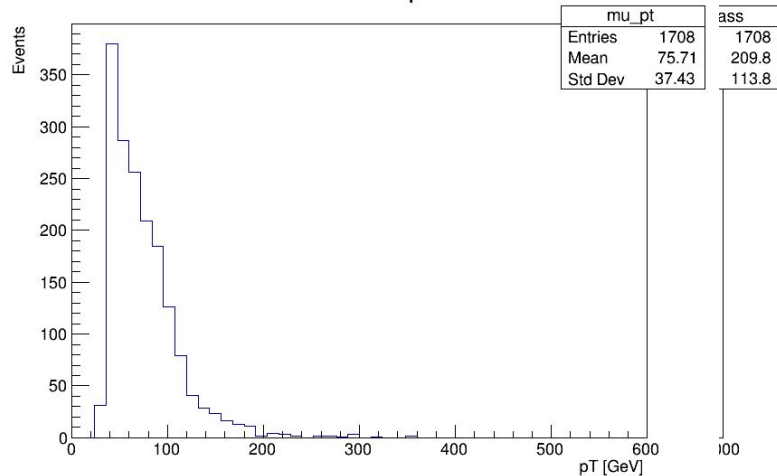
Data: Tau Eta



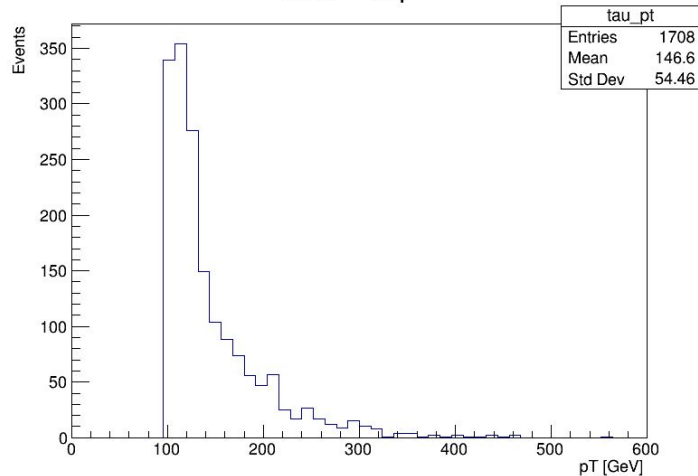
Data: Muon Eta



Data: Muon pT



Data: Tau pT



06/11/2025

data	
1 tau. 1 muon	
✓	
✓	
>63	
>7	
>1	
>100	
>=3	
>35	
>0.4	
<2.4	
opposite	

Sample details: Data with these cuts applied

02

Variable plots

Signal